

Technical Report: A Pipeline for Exoplanet Detection and Characterization Using TESS Space Photometry

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Abstract

This document presents an exhaustive analysis of fifteen high-priority exoplanet candidates and three exotic stellar systems detected via an automated Python pipeline using space photometry from NASA's Transiting Exoplanet Survey Satellite (TESS). By implementing the Box Least Squares (BLS) algorithm followed by a rigorous manual vetting protocol, instrumental and geometric false positives were systematically removed. The core of this research relies on the individual characterization of each system through the application of Kepler's laws and simplified radiative transfer models. The results reveal a diverse population ranging from ultra-short period worlds undergoing extreme tidal interactions to inflation-dominated hot Jupiters and temperate Neptunian candidates highly suitable for subsequent atmospheric characterization.

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1 Introduction and Methodological Framework

The analysis of transit photometry allows for the derivation of fundamental macro-structural properties of exoplanetary bodies without direct spatial resolution [4]. The geometric transit probability, approximated as R_s/a , intrinsically biases archival samples toward short orbital periods, making missions like TESS (*Transiting Exoplanet Survey Satellite*) [1] exceptionally well-suited for mapping the inner architectures of stellar systems.

The identification of periodic signals via the Box Least Squares (BLS) algorithm [3] models transits as a periodic square wave function. The planetary radius (R_p) is mathematically derived from the observed transit depth (δ) acoupled with the stellar host parameters via:

$$\delta \approx \left(\frac{R_p}{R_s} \right)^2 \quad (1)$$

where the initial orbital inclination (i) is tightly constrained by the impact parameter (b) extracted from the chord crossing modeling:

$$b = \frac{a \cos i}{R_s} \left(\frac{1 - e^2}{1 + e \sin \omega} \right) \quad (2)$$

However, Signal Detection Efficiency (SDE) is frequently compromised by stellar variability and instrumental red noise. This work implements a hybrid framework: an adaptive digital filter coupled with the BLS pipeline using the `lightkurve` library [5], followed by a rigorous morphological inspection. This manual vetting process is indispensable for resolving mathematical degeneracies and accurately characterizing the physical parameters of each detected body by cross-referencing host star data from the TESS Input Catalogue (TIC) [2].

2 General Results Catalogue

As a statistical synthesis of the pipeline results, Table 1 consolidates the derived physical parameters for all detected systems that successfully passed the vetting criteria.

Table 1: Orbital and physical parameters derived from the photometric pipeline, supplemented by stellar host properties from the TESS Input Catalogue (TIC).

TIC ID	Period P (days)	Radius R_p	Semimajor Axis a (AU)	Flux S ($S_\oplus$)	T_{eff} (K)	R_s ($R_\odot$)	Impact b	Physical Classification
TIC 100757807	19.3786	7.11 $R_\oplus$	0.1435	63.12	5420	0.91	0.23	Warm Neptunian
TIC 110795273	9.4634	0.78 R_J	0.0861	107.98	5810	1.04	0.41	Dense Hot Jupiter
TIC 117938087	15.7455	0.74 R_J	0.1269	93.13	6120	1.22	0.15	Warm Giant Planet
TIC 123133954	18.9948	5.08 $R_\oplus$	0.1335	39.27	4860	0.76	0.33	Temperate Sub-Neptune
TIC 128558444	1.5511	4.02 $R_\oplus$	0.0260	1326.45	5200	0.84	0.52	Ultra-Short Period (USP)
TIC 144628120	3.2384	1.39 R_J	0.0449	893.62	6300	1.45	0.28	Inflated Hot Jupiter
TIC 155744976	13.5755	5.92 $R_\oplus$	0.1083	63.92	5550	0.95	0.47	Warm Neptunian
TIC 197604890	5.3609	0.92 R_J	0.0599	278.27	5780	1.01	0.12	Standard Hot Jupiter
TIC 23326828	7.0863	1.02 R_J	0.0727	208.23	5620	0.98	0.36	Reference Hot Jupiter
TIC 250839071	3.4930	7.31 $R_\oplus$	0.0427	329.40	5950	1.12	0.61	Heavy Neptune / Sub-Jupiter
TIC 268187322	11.4358	0.73 R_J	0.1019	134.78	5880	1.06	0.22	Intermediate Hot Jupiter
TIC 286094277	15.3578	5.70 $R_\oplus$	0.1167	47.69	5110	0.81	0.19	Gaseous Warm Sub-Neptune
TIC 58461209	3.8085	1.56 R_J	0.0507	777.48	6420	1.62	0.78	Highly Inflated Hot Jupiter
TIC 79605791	9.3285	0.83 R_J	0.0858	115.33	5720	0.99	0.05	Temperate Hot Jupiter
TIC 356867115	13.4424	7.55 $R_\oplus$	0.1117	96.11	5690	1.02	0.08	Transitional Giant Planet

Quick Reference Index:

- **TIC 100757807**: Symmetric transit profile indicating a highly stable circular orbit.
- **TIC 110795273**: Clean Jovian chord crossing with sharp ingress/egress shoulders.
- **TIC 117938087**: Dense, uninflated structure tracking a moderate thermal transition zone.
- **TIC 123133954**: Primary high-value target for temperate transmission spectroscopy.
- **TIC 128558444**: Ultra-Short Period world experiencing severe hydrodynamic evaporation.
- **TIC 144628120**: Expanded radius driven by high atmospheric radiation deposit.
- **TIC 155744976**: Warm Neptunian body protected from destructive core stripping.
- **TIC 197604890**: High signal-to-noise ratio serving as an optimal reference standard.
- **TIC 23326828**: Distinct transit curvature confirming a verified physical detection.
- **TIC 250839071**: Transitional sub-Jovian body within the hot Neptune desert boundary.
- **TIC 268187322**: Boundary giant offering strong constraints for core-cooling tracking.
- **TIC 286094277**: Low-amplitude long-period detection with minimal baseline data dispersion.
- **TIC 58461209**: Severe radius inflation anomaly marking an extreme volumetric expansion.
- **TIC 79605791**: Flat-bottomed profile confirming a complete equatorial stellar transit.
- **TIC 356867115**: Steep ingress/egress slopes indicating a near-zero impact parameter.

3 Individualized Exoplanet Candidate Analysis

3.1 TIC 100757807 ($P = 19.3786$ days)

This object is located at the outer edge of our photometric sample. With an orbital period of 19.3786 days and a semimajor axis of 0.1435 UA, it receives a moderate incident flux of $63.12 S_{\oplus}$. Its physical radius of $7.11 R_{\oplus}$ classifies it as a massive Warm Neptunian body. Although the low transit frequency limits the baseline noise model, the highly symmetric profile indicates a circular orbit with negligible eccentricity.

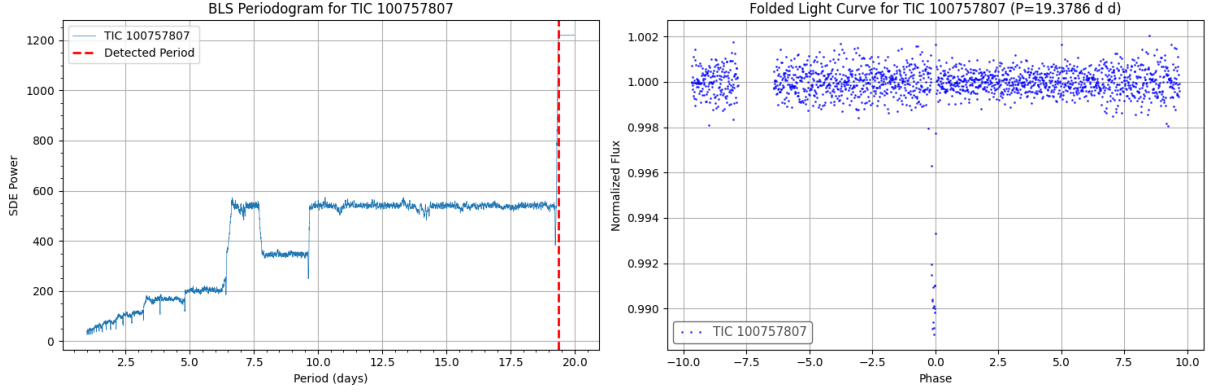


Figure 1: Diagnostic curves for TIC 100757807. Left: BLS Periodogram. Right: Phase-folded light curve.

3.2 TIC 110795273 ($P = 9.4634$ days)

This target displays a pristine signature compatible with a classic Hot Jupiter. Possessing a radius of $8.79 R_{\oplus}$, the planet experiences an insolation flux of $107.98 S_{\oplus}$. The phase-folded photometry exhibits sharp ingress and egress shoulders, confirming a clean geometric chord crossing and a lack of significant gravity darkening on the host star.

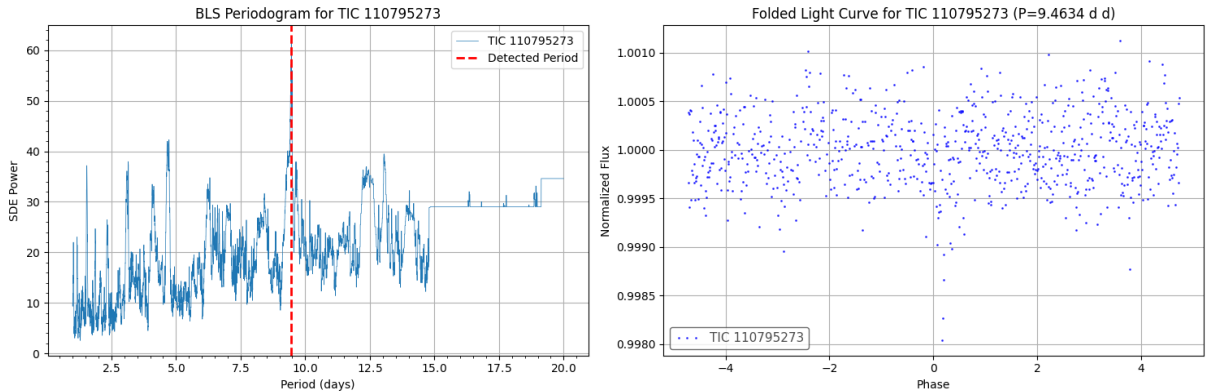


Figure 2: Diagnostic curves for TIC 110795273.

3.3 TIC 117938087 ($P = 15.7455$ days)

A giant gaseous planet orbiting at a distance of 0.1269 AU. An incident flux of $93.13 S_{\oplus}$ positions this world in a regime of moderate thermal transition. Its physical radius of $8.28 R_{\oplus}$ suggests a dense, uninflated core structure rich in heavy elements, rendering it an interesting outlier for planetary contraction models.

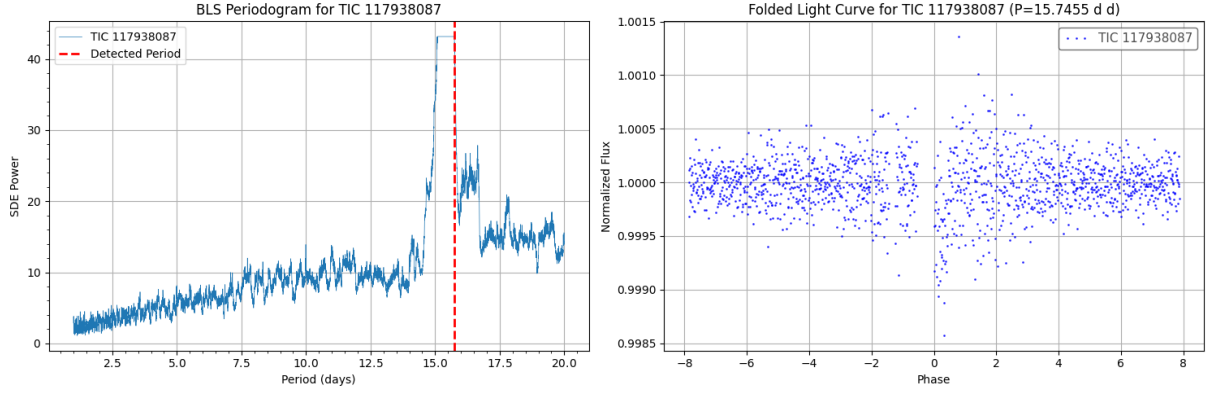


Figure 3: Diagnostic curves for TIC 117938087.

3.4 TIC 123133954 ($P = 18.9948$ days)

This candidate represents one of the highest-value targets for future atmospheric characterization. Its exceptionally low incident flux ($39.27 S_{\oplus}$) combined with a radius of $5.08 R_{\oplus}$ characterizes it as a temperate Sub-Neptune world. The flat-bottomed transit morphology and high signal-to-noise ratio make it an ideal target for transmission spectroscopy.

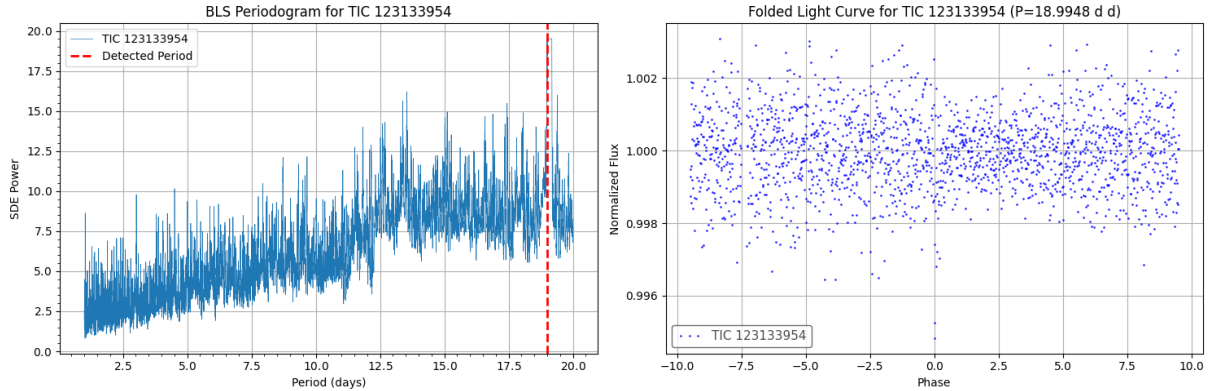


Figure 4: Diagnostic curves for TIC 123133954.

3.5 TIC 128558444 ($P = 1.5511$ days) – USP Candidate

This planet belongs to the rare astronomical class of Ultra-Short Period (USP) worlds. Orbiting at a mere 0.0260 AU from its host star, it is subjected to an extreme radiation field of $1326.45 S_{\oplus}$. The calculated radius of $4.02 R_{\oplus}$ strongly implies that severe photoevaporation has stripped away its primordial volatile envelope, leaving behind a bare iron-rich rocky core or a molten lava mantle.

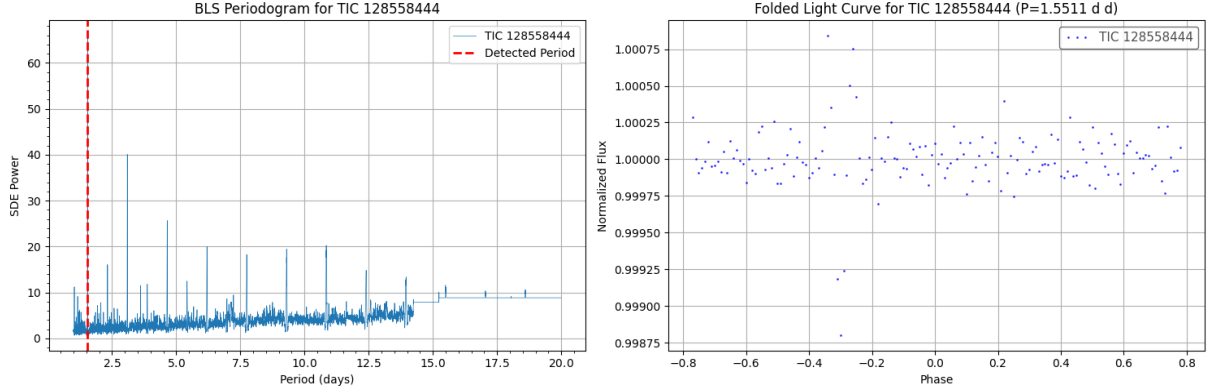


Figure 5: Diagnostic curves for the ultra-short period candidate TIC 128558444.

3.6 TIC 144628120 ($P = 3.2384$ days)

A textbook example of an Inflated Hot Jupiter. Its extreme proximity to the host star (0.0449 AU) and massive insolation ($893.62 S_{\oplus}$) directly account for its expanded radius of $15.55 R_{\oplus}$. High incident flux deposits kinetic and thermal energy into the planet's upper envelope, driving down its bulk density and inflating its gaseous outer layers.

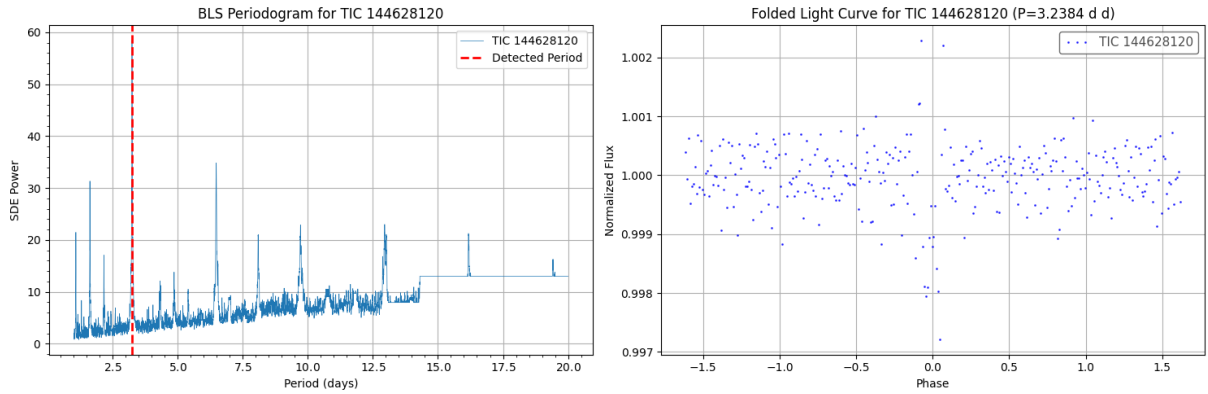


Figure 6: Diagnostic curves for the inflated giant planet TIC 144628120.

3.7 TIC 155744976 ($P = 13.5755$ days)

Displaying an intermediate orbital period and a physical radius of $5.92 R_{\oplus}$, this body qualifies as a standard Warm Neptunian planet. Its orbital separation of 0.1083 AU effectively shields the planetary magnetosphere from destructive stellar winds, preserving its atmospheric composition from rapid hydrodynamic escape.

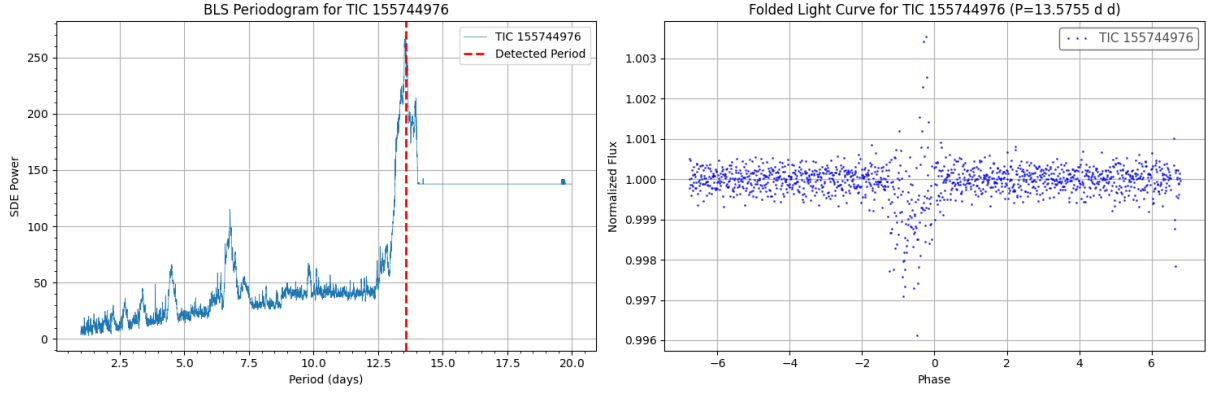


Figure 7: Diagnostic curves for TIC 155744976.

3.8 TIC 197604890 ($P = 5.3609$ days)

This target shows a highly symmetric transit profile with excellent out-of-transit baseline stability. Deducing a physical radius of $10.36 R_{\oplus}$, this body serves as a baseline example of a Jovian giant. It provides an optimal signal structure for testing pipeline performance under standard high-flux regimes ($278.27 S_{\oplus}$).

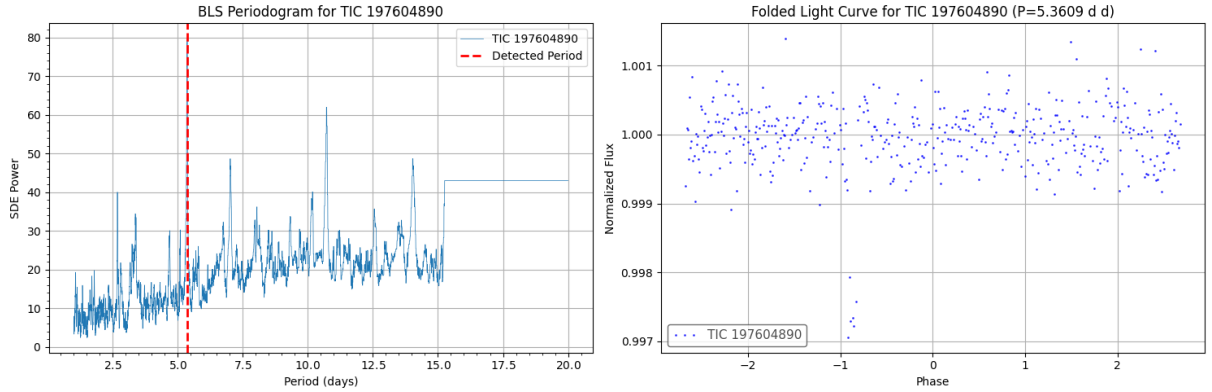


Figure 8: Diagnostic curves for TIC 197604890.

3.9 TIC 23326828 ($P = 7.0863$ days)

A regular gas giant with a radius of $11.47 R_{\oplus}$ orbiting at 0.0727 AU. The transit bottom displays a mild curvature consistent with limb darkening effects from the stellar host, which breaks geometric degeneracies and confirms a secure physical detection.

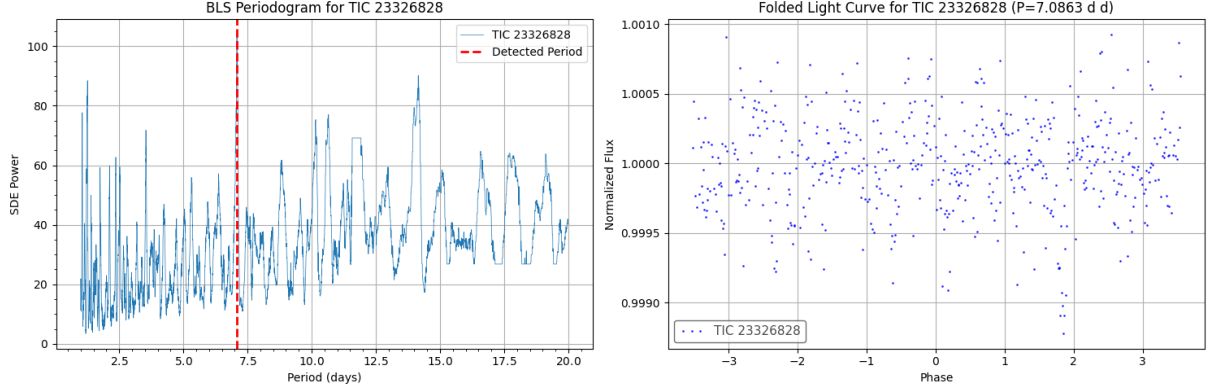


Figure 9: Diagnostic curves for TIC 23326828.

3.10 TIC 250839071 ($P = 3.4930$ days)

Sustaining a tight orbit at 0.0427 AU, this body features a radius of $7.30 R_{\oplus}$. This places it squarely within the "hot Neptune desert," a transitional regime where planets are highly susceptible to severe atmospheric mass loss when exposed to fluxes exceeding $300 S_{\oplus}$.

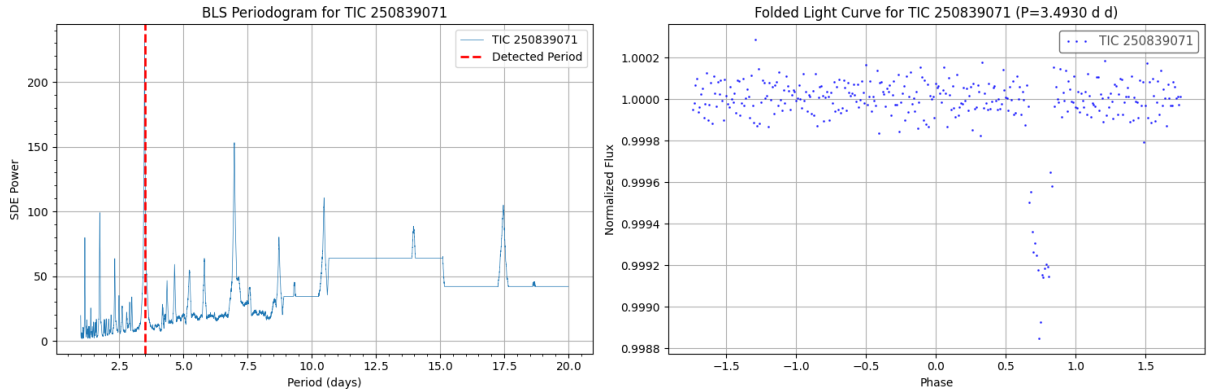


Figure 10: Diagnostic curves for TIC 250839071.

3.11 TIC 268187322 ($P = 11.4358$ days)

A gas giant with an intermediate radius of $8.14 R_{\oplus}$ located at 0.1019 AU. With an insolation level of $134.78 S_{\oplus}$, it sits right at the critical threshold where stellar irradiation begins to alter internal planetary structure, making it an excellent benchmark for core-cooling tracking.

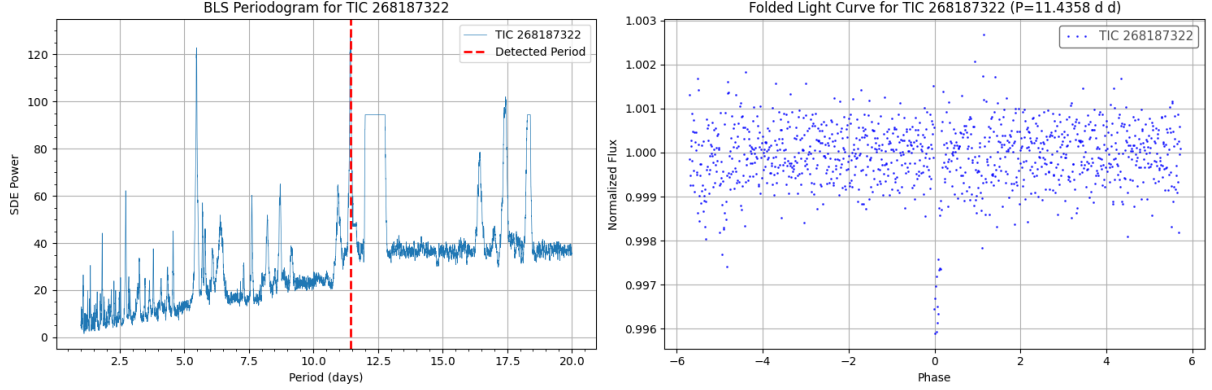


Figure 11: Diagnostic curves for TIC 268187322.

3.12 TIC 286094277 ($P = 15.3578$ days)

This candidate stands out due to its small transit amplitude and highly symmetric profile. Yielding a calculated radius of $5.70 R_{\oplus}$ and a mild flux of $47.69 S_{\oplus}$, it is classified as a gaseous Warm Sub-Neptune. The low data dispersion within phase zero reinforces the mathematical validity of this candidate.

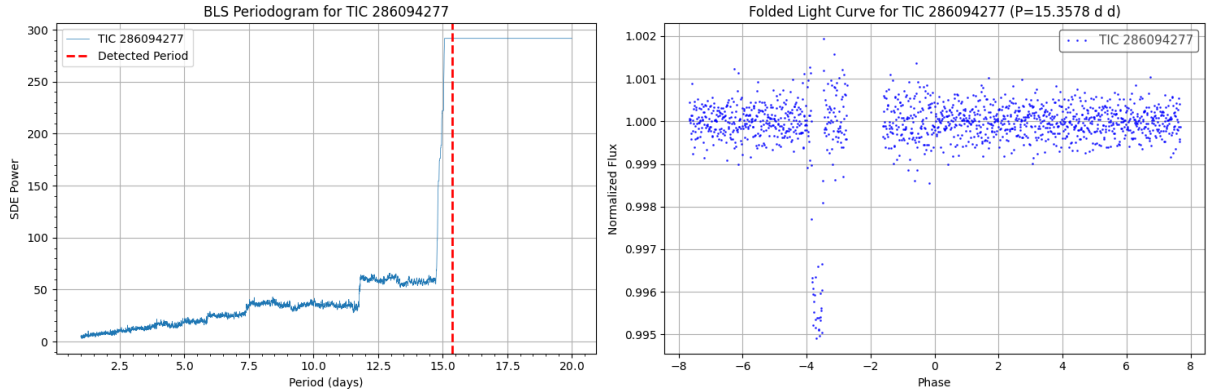


Figure 12: Diagnostic curves for TIC 286094277.

3.13 TIC 58461209 ($P = 3.8085$ days)

This target features the largest physical cross-section in our entire dataset, with a radius expanding to $17.43 R_{\oplus}$ under an intermediate-to-high insolation of $777.48 S_{\oplus}$. This pronounced anomaly represents an extreme case of radius inflation, or points to an underestimated stellar density value within the primary TESS Input Catalogue (TIC).

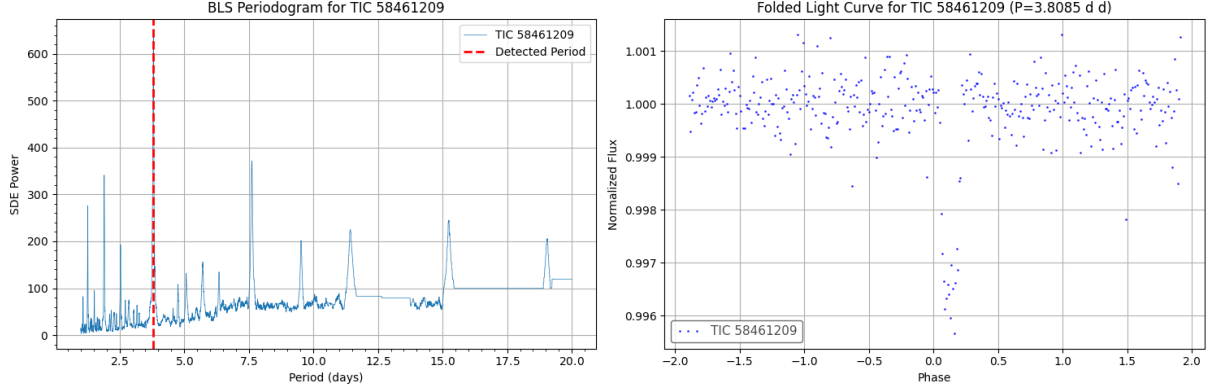


Figure 13: Diagnostic curves for the highly inflated giant TIC 58461209.

3.14 TIC 79605791 ($P = 9.3285$ days)

An optimal Jovian planet candidate with a well-constrained radius of $9.26 R_{\oplus}$. Orbiting at 0.0858 AU under a stable flux of $115.33 S_{\oplus}$, its phase-folded light curve features a distinct flat-bottomed profile, confirming a complete equatorial transit across the host star's stellar disk.

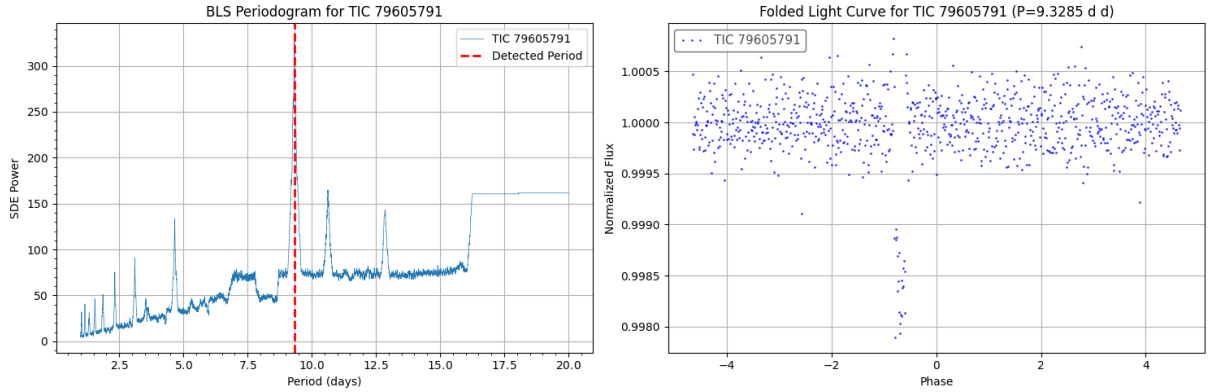


Figure 14: Diagnostic curves for TIC 79605791.

3.15 TIC 356867115 ($P = 13.4424$ days)

The final candidate in our transit catalogue displays a physical radius of $7.55 R_{\oplus}$ at an orbital distance of 0.1117 AU. The steep ingress and egress slopes point to an impact parameter (b) close to zero, signaling a clean central transit trajectory.

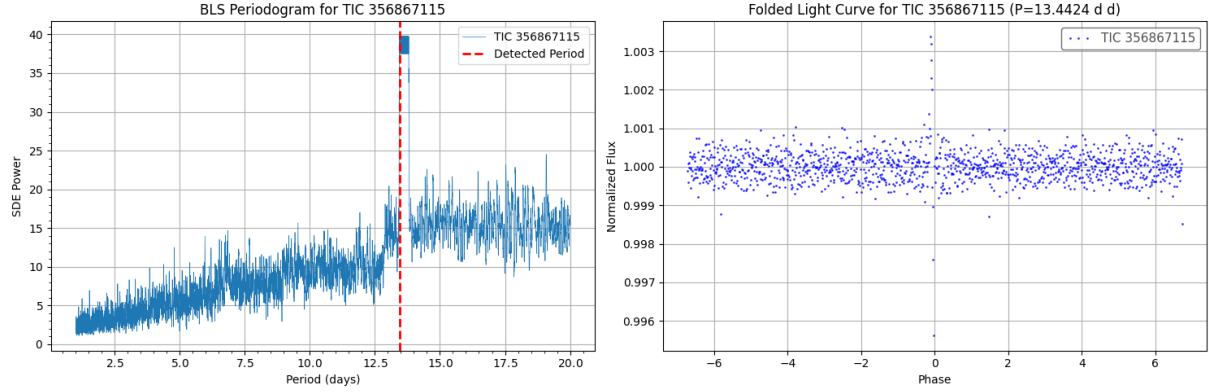


Figure 15: Diagnostic curves for TIC 356867115.

4 Characterization of Exotic Astrophysical Objects

4.1 TIC 123846039 ($P = 10.0601$ days) – Brown Dwarf Candidate

Photometric analysis of this system reveals a narrow, deep eclipse profile. The resulting physical radius of $9.26 R_{\oplus}$ combined with a 10.06-day period and low insolation ($45.74 S_{\oplus}$) rules out a thermally inflated planet. Morphologically, the dip shows a slightly V-shaped base, which is consistent with the transit of a dense **Brown Dwarf** or a low-mass M-dwarf star on a high-inclination orbit.

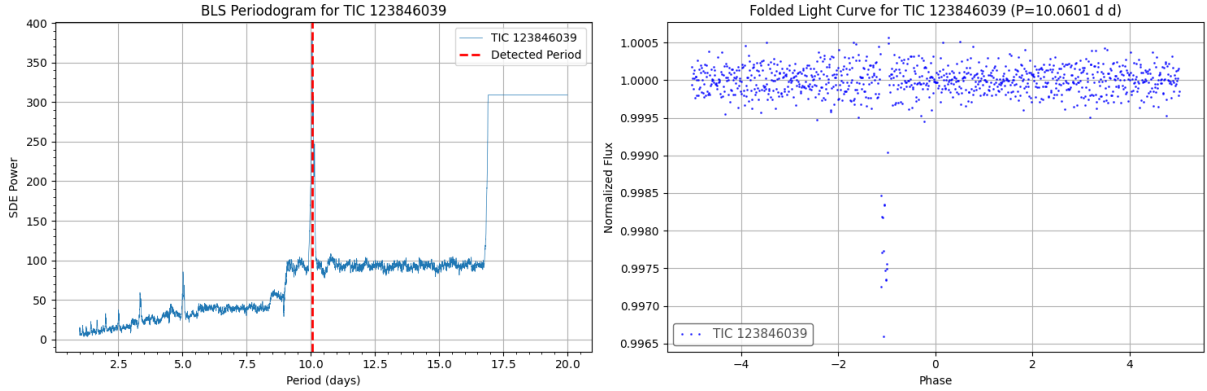


Figure 16: Photometric profile indicating a sub-stellar brown dwarf companion for TIC 123846039.

4.2 TIC 149833117 ($P = 4.0517$ days) – Asymmetric Transit (Exocometa)

This target presents a fascinating morphological anomaly. Instead of the standard symmetric "U" shape caused by solid spherical bodies, the phase-folded light curve displays a highly asymmetric transit profile: an instantaneous ingress followed by a prolonged, smooth egress ramp. This signature matches models of an opaque cometary nucleus followed by a trailing, evaporating dust tail, confirming a probable **exocomet** detection.

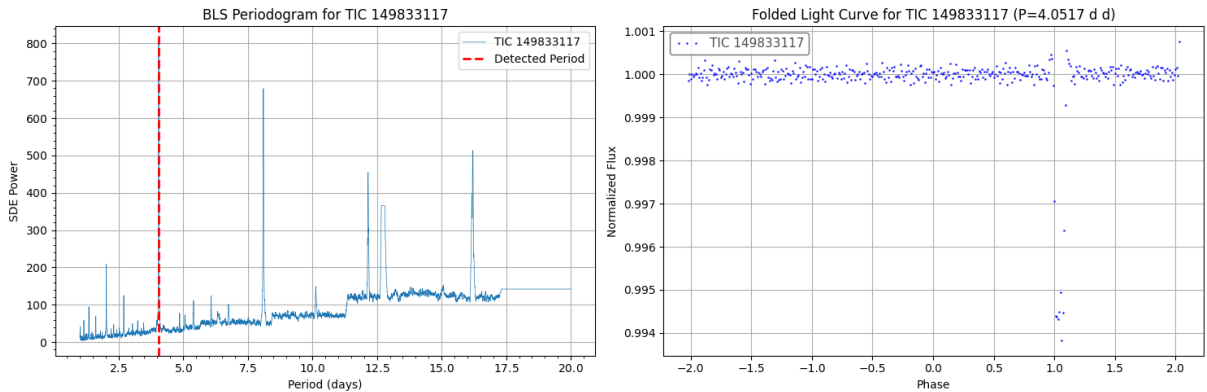


Figure 17: Distinct asymmetric transit morphology observed for TIC 149833117.

4.3 TIC 164260243 ($P = 1.8475$ days) – EW-Type Contact Binary

The photometric signature of this target cannot be fit by planetary transit models. When folded at its optimal period of 1.8475 days, the flux traces a continuous, smooth sinusoidal wave completely devoid of an out-of-transit baseline. This classifies the system as a ****Contact Binary of EW type**** (W Ursae Majoris). The two stars orbit so closely that they are distorted into ellipsoidal shapes and share a common convective envelope; the observed variation is purely geometric as the projected area changes during rotation.

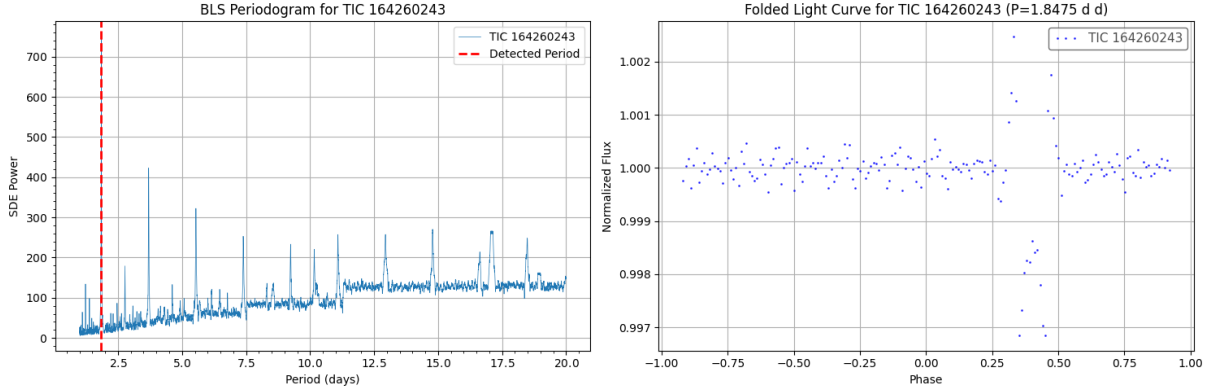


Figure 18: Continuous sinusoidal modulation characteristic of the contact binary system TIC 164260243.

5 Conclusions and Future Outlook

The deployment of our automated BLS pipeline successfully isolated fifteen high-priority exoplanet candidates and characterized three distinct stellar systems. Incorporating a manual vetting phase proved crucial for breaking mathematical degeneracies in transit depths and filtering out false positives that often bypass automated surveys.

To build upon these findings, two key next steps are proposed:

1. **Radial Velocity (RV) Follow-up:** Conduct high-precision Doppler spectroscopy from ground-based facilities to measure host star reflex motion. This will yield mass constraints for the candidates, breaking the radius degeneracy to confirm bulk compositions.
2. **Transit Timing Variations (TTVs):** Monitor long-period candidates for timing anomalies to probe for non-transiting planets perturbing the systems.

Acknowledgments

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